LESSON 3.2

Understand Rates and Unit Rates

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can describe a rate using its two different units and use a model to find a travel time, and I can find a unit rate to compare prices and decide the better buy.

WHY THIS MATTERS You're comparing ticket prices at different game booths in the arcade. Booth A charges $3 for 5 games, and Booth B charges $5 for 8 games. Which booth gives you a better deal? To find out, you need to figure out the cost per game — that's the unit rate!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Rates and Unit Rates *Spanish: Tasas y tasas unitarias*	A rate compares different units; a unit rate compares a quantity to 1 unit.	180 miles in 3 hours = 60 miles per 1 hour.
Rate *Spanish: Tasa*	A ratio comparing two amounts with different units, like miles per hour.	96 pages / 3 hours
Unit Rate *Spanish: Tasa unitaria*	A rate for just 1 of something, like cost for 1 item.	32 pages per 1 hour
Per *Spanish: Por*	For each one. Example: 5 dollars per book.	miles per hour, cost per pound
Ratio *Spanish: Razón*	A way to compare two amounts, like 3 to 2.	5 to 3 or 5:3
Unit price *Spanish: Precio unitario*	A unit rate that gives the cost of exactly 1 item, like $0.60 per game.	$3 for 5 games → $0.60 per 1 game

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A printer prints 90 pages in 6 minutes at a steady rate. How many pages in 7 minutes? How long for 45 pages?

1. Divide: 90 ÷ 6 = 15 pages per minute.
2. 7 minutes: 15 × 7 = 105 pages.
3. 45 pages: 45 ÷ 15 = 3 minutes.

Answer: 105 pages; 3 minutes

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A booth charges $4 for 8 games. What do we divide?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The rate is _____ for every _____ .
- I divided _____ by _____ to find the unit rate.
- The unit rate is _____ per _____ .
- To find the amount for _____ , I _____ .

Word bank rate | unit rate | per | each | divide | multiply | for every | speed | predict | constant | hour | pound

WATCH OUT!

A common mistake

A common mistake in Rates and Unit Rates is comparing total costs instead of unit rates: a student sees Pack X at $4.00 for 10 tokens and Pack Y at $6.00 for 16 tokens and picks Pack X just because $4.00 is the smaller number, without dividing to find the cost per token ($0.40 for Pack X vs. $0.375 for Pack Y

LESSON 3.2

Understand Rates and Unit Rates

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

A booth charges $4.50 for 9 games. What is the unit rate?

(A) $0.45 per game

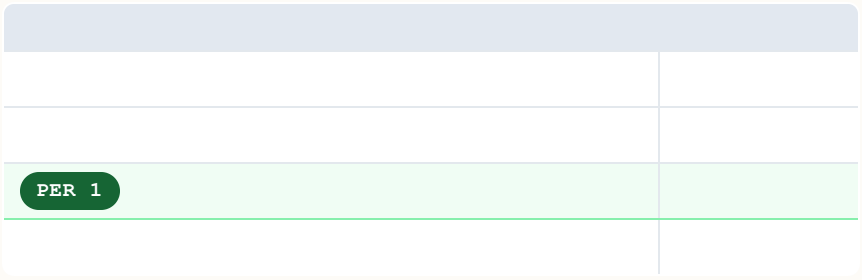
(B) $0.50 per game

(C) $2.00 per game

(D) $4.50 per game

STEPS TO TRY

1. You know the total cost ($4.50) and the total games (9). A unit rate tells the cost of just ONE game.
2. To find a unit rate, divide the total cost by the number of games so the second quantity becomes 1.
3. Set up the division: $4.50 ÷ 9. Think of $4.50 as 450 cents if that makes the division easier.



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

2 **Circle the letter of the best answer. Show how you know.**

A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

(A) 6 miles per gallon

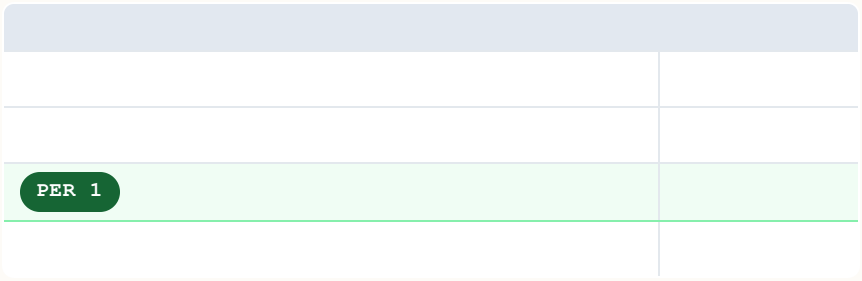
(B) 30 miles per gallon

(C) 36 miles per gallon

(D) 180 miles per gallon

STEPS TO TRY

1. Notice the units in the question: miles PER gallon. That tells you miles go on top and gallons on the bottom.
2. Divide the total miles by the total gallons so you get the miles for just 1 gallon.
3. Start the division: 180 ÷ 6. Think: 6 × what number lands on 180?



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2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

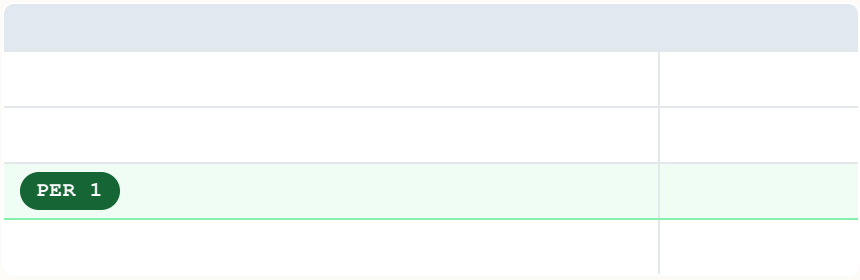
3 **Circle the letter of the best answer. Show how you know.**

A pack of 8 markers costs $6.00. What is the cost per marker?

(A) $0.60
(B) $0.75
(C) $0.80
(D) $1.33

STEPS TO TRY

1. You have 8 markers costing $6.00 total. 'Cost per marker' means the price of ONE marker.
2. Divide the total price by the number of markers — cost goes first, markers second.
3. Set up $6.00 ÷ 8. Tip: $6.00 is 600 cents, so try 600 ÷ 8 cents.



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

4 **Circle the letter of the best answer. Show how you know.**

Which is a unit rate?

(A) 12 miles per 1 hour
(B) 24 miles in 2 hours
(C) 36 miles for 3 gallons
(D) 100 points in 5 games

STEPS TO TRY

1. A unit rate always compares to exactly ONE of something — look at the second number in each choice.
2. Check each choice: is the second quantity (hours, gallons, games) equal to 1? Only one choice passes that test.
3. Start with '24 miles in 2 hours' — the second quantity is 2, not 1, so that one is not a unit rate. Test the rest the same way.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How does finding the unit rate help you compare the booths?

SENTENCE STARTER The best deal is Booth ___ because its unit rate of ___ per game is the ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help